

Simulation of recovery of pre-disturbance size structure in populations of *Porites* spp. damaged by the crown of thorns starfish *Acanthaster planci*

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Abstract

Outbreaks of the crown of thorns starfish *Acanthaster planci* (L.) have caused high levels of mortality and injury in corals on the Great Barrier Reef. In surveys conducted in 1985 and 1986, it was estimated that a quarter of the massive *Porites* spp. corals surveyed at five reefs – John Brewer, Rib, Potter, Feather and Green Island – had been killed outright. In addition, there was minor to severe injury to colonies throughout the coral's size range. A population model was used to evaluate this damage. The evaluations were based on simulations of the time necessary for prior abundances of large colonies (>2 m diam) to be re-established and on simulations of the number of additional disturbances the populations could withstand before becoming locally extinct. The affect of recurrent disturbances on populations receiving recruits was also simulated. Assuming no further disturbance, the model predicts minimum recovery times in excess of 50 yr for most of the populations, and 9 to 100 yr for sub-populations within reefs. For populations subjected to repeated disturbance every 10 to 30 yr, it predicted that all large colonies would soon be lost unless there was both low background mortality and a major recruitment of *Porites* spp. between outbreaks. In the worst-case scenario of failed recruitment and high background mortality, four of the five populations could withstand as few as two to three additional disturbances equivalent to those earlier in this decade.

Introduction

The damage to coral populations caused by outbreaks of the crown of thorns starfish *Acanthaster planci* (L.) varies from place to place. It ranges from minor and patchy injury and mortality in corals (e.g. Porter 1972), to a fundamental change in reef state; from dominance by living animals (corals), to dominance by dead and injured animals (coral

skeletons), to dominance by plants (Endean 1973, Pearson 1981, Cameron and Endean 1985, Done 1985). The regenerative capabilities of corals can be considerable, through recolonization and through repair (Bak and Steward-Van Es 1980, Pearson 1981, Hughes and Jackson 1985). However, as these responses are neither instantaneous nor infinite, the affected area will carry a reduced cover of live coral for some time. While the time for restitution of a high cover of fast-growing corals may in some places be measured in years to a decade or so (Pearson 1981, Done 1985), the replacement of the oldest individuals of slow-growing species can take up to several centuries (Cameron and Endean 1985).

Although these processes are well understood in general terms, quantitative evaluation of disturbance and estimation of recovery times at specific places have not previously been attempted; biologists do not generally have sufficient site- or species-specific information. In particular, the future recruitment to, and survival of, a recovering coral population are unknown. The present paper shows how field data consisting of a categorization of individual colonies by their size and by their amount of injury (including death), may be evaluated using a matrix model. Using best- and worst-case assumptions about the coral population's future performance, the model provides a range of estimates for the time for replacement of pre-disturbance population structures, and for the number of additional disturbances which could take a local population to extinction.

The genus *Porites* represents an extreme in the spectrum of life-history strategies found in corals. Individual heads grow slowly (about 1 cm yr⁻¹ radially; Isdale 1983), reach large size (several meters diameter and height; Fig. 1) and old age (several centuries). While they spawn or planulate annually (Kojis and Quinn 1981), recruitment from planular larvae appears to be rare and possibly secondary in importance to recruitment from fragmentation in some habitats (Highsmith 1980, Kojis and Quinn 1981, Potts et al. 1985). They occur in mixed communities, mostly comprising faster growing species, and their population densities are patchy, tending to be low on reefs affected by crown of thorns

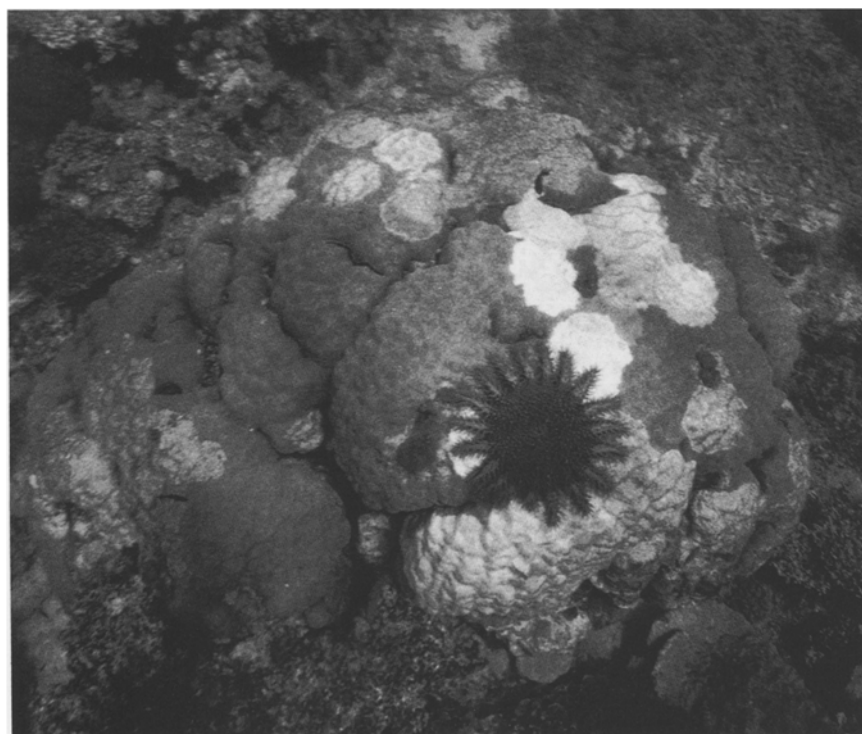


Fig. 1. *Porites lobata* and *Acanthaster planci*. Massive coral with its predator, the crown of thorns starfish. White areas of colony are the current day's feeding scars; starfish is approx 350 mm in diameter

Table 1. Details of reefs included in study

Reef	Outbreak history	Source
John Brewer	1969 1983	Brown and Miller (1972) Done (1985)
Rib	1966 1982	Endean and Stablum (1975) GBRMPA data base ^a
Potter	?1966 1984	— GBRMPA data base ^a
Feather	1966 1981	Pearson and Endean (1969) GBRMPA data base ^a
Green Island	1962 1979	Barnes (1966) Endean (1982)

^a Unpublished records of Great Barrier Reef Marine Park Authority

starfish (Done 1982, 1985). They are frequently injured or killed during crown of thorns outbreaks (Done 1985, Endean and Cameron 1985), but they continue skeletal deposition in the live areas of the colony which remain (Done 1987).

Cameron and Endean (1985) noted that the recovery time for populations which are entirely destroyed is equal to the age of the oldest coral, in some cases several centuries. However, in populations which are not completely destroyed, the recovery time will be less (Done 1987). In such cases, much recovery is through growth and repair of the survivors, and estimation of recovery time requires both an assessment of the extent of injury among different sizes of corals, a projection of growth and regeneration, and allowance for the contribution made by new recruits to the population.

Materials and methods

Field data

Population densities and the extent of injury and mortality in *Porites* spp. were assessed at five reefs following the most recent *Acanthaster planci* (L.) outbreaks during 1979–1984 (Table 1). These reefs – John Brewer, Rib, Potter, Feather and Green Island – are mid-shelf reefs scattered over a 300 km section of the central Great Barrier Reef. All were severely disturbed (Crown of Thorns Study 1986a) and all (probably including Potter, for which only anecdotal data are available) were affected during the 1960's (Table 1).

The *Porites* spp. surveys at four reefs were conducted in 1985 and 1986 by the author and members of the Crown of Thorns Commonwealth Community Employment Team (Crown of Thorns Study 1986b). We used belt transects (50 m × 0.6 m for colonies <0.6 m diam, and approx 200 m × 20 m for colonies >0.6 m diam) at 4 to 12 sites around the perimeter of the reefs (Fig. 2; and Done 1985, 1987). The fifth reef, John Brewer Reef, had already been surveyed using methods which were identical except for the width of the 50 m belt transect, which was 0.3 m (Done 1987). Location of the sites was dictated by accessibility, with an attempt to survey regularly around the reef perimeters. On simple reef slopes, a fibreglass tape measure laid along three depth contours (approximately 3, 6 and 9 to 12 m) was used to measure the 50 m transects. Where the topography was flatter, the three 50 m transects were placed a few metres apart at a similar depth to each other. Each colony within a belt transect was scored into one of five size categories and four injury categories (Table 2). Only deaths and injuries attributable to the most recent starfish outbreak were recorded, these being determined by a level of fouling and/or

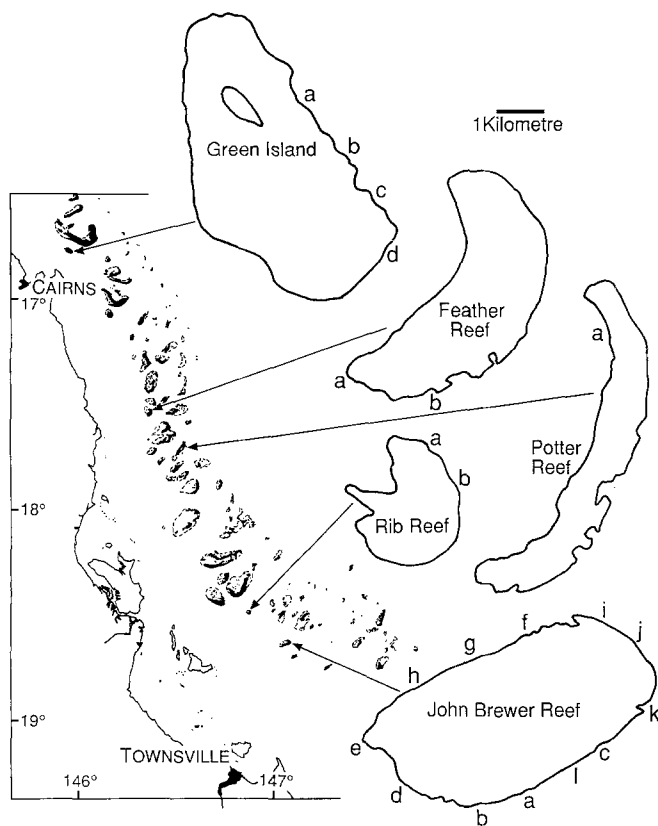


Fig. 2. Locations of study reefs and sectors (a–l) on study reefs

a distance below the adjacent living surface which was consistent with the time since the last outbreak. The populations contained a mixture of three species which are indistinguishable except at the microstructure level: *P. lutea* Edwards and Haime, *P. lobata* Dana, and *P. australiensis* Vaughan (Veron and Pichon 1982).

The data are considered on whole reef (pooled data) and individual sector (or local population) bases. At John Brewer Reef, where the positions of the surveys were accurately fixed using compass bearings, individual sectors consisted of a single set of one 200 m belt transect and three 50 m belt transects sampled from within the 200 m belt transect. At the remaining reefs, the location of the survey areas was identified only to a broad general area. Poor weather at the time of the second survey made it impossible to centre the 50 and 200 m transects at the same point on the reef perimeter. Here, multiples of three 50 m and of one 200 m belt-transects were used to estimate the population over larger sectors (Fig. 2; Table 3). All density estimates are expressed as the number of starfish in each size class per hectare, using conversion factors of 111.0 (for each set of three 50 m $\times$ 0.6 m belt transects), 222.0 (for each set of three 50 m $\times$ 0.3 m belt transects) and 2.5 (for each 200 m $\times$ 20 m belt transect).

Matrix model for injury, regrowth and population growth

Injury causes a reduction in live surface of the colony, or “shrinkage” (Done 1987, after Hughes 1984). Subsequently,

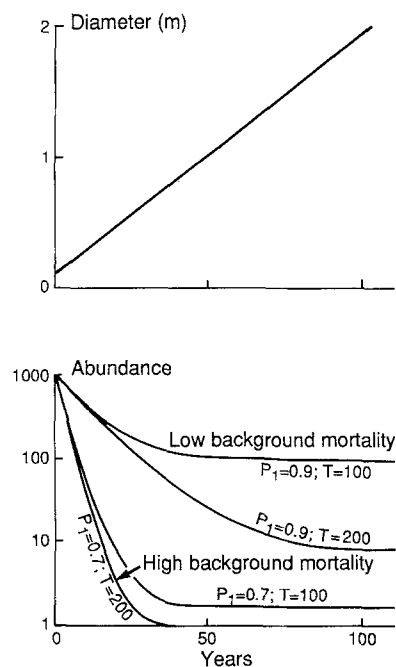


Fig. 3. *Porites* spp. Diameter and abundance (no. ha⁻¹) of a cohort of colonies recruiting at a diameter of 0.1 m and growing while subject to four simulated background mortalities. Curves were simulated using transition matrices based on Eqs. (1) and (2), with values of P_1 (initial annual survival probability) of 0.7 and 0.9 and T (colony escape-threshold diameter for background mortality) of 100 and 200 cm, respectively

uninjured portions continue to grow, and frequently adjacent living areas will fuse. The population dynamics of this cycle of injury and recovery may be described using two transition matrices, one describing changes between disturbances and one the changes during a disturbance (Hughes 1984). Transition matrices define the probability that colonies in a particular size class will grow into a larger size class, remain in the same size class, die, or shrink into a smaller size class. Death and shrinkage become the predominant changes during periods with severe disturbance. In closed populations, recruitment of individuals to the smallest size class may also be defined in the matrix (Usher 1972).

In this study, the normal, “non-outbreak” population dynamics of *Porites* spp. is assumed to be simply the net result of individual growth, background mortality and, in some simulations, recruitment of new corals. Background mortality includes all causes of mortality and injury in non-outbreak years, including low-level predation by residual starfish populations, and overgrowth by faster growing benthos. The non-outbreak matrix divides the population into 2 cm increments in diameter, and with an annual diameter growth rate also equal to 2 cm, the annual probability ($a_{j,j+1}$) of a coral growing into its next size class is:

$$a_{j,j+1} = P_j, \quad (1)$$

where $P_j = P_1 + (j \times i)$, and P_j = the probability of survival while in the j th size class, P_1 = the probability of survival of a 0.1 m coral, and i = the constant increment in P_j per incre-

Table 2. *Porites* spp. Percentage of colonies categorized as dead (a) or as possessing the following proportions of living surface area: b, <0.33; c, 0.33 to 0.67; d, >0.67, following damage by *Acanthaster planci*. Locations of sectors are shown in Fig. 2

Location	Size class (diam range) and injury category																				Total colonies					No. of transects (50 m, 200 m)	
	I (0.1–0.29 m)				II (0.3–0.59 m)				III (0.6–0.99 m)				IV (1.0–1.99 m)				V (≥ 2.0 m)				I	II	III	IV	V		
	a	b	c	d	a	b	c	d	a	b	c	d	a	b	c	d	a	b	c	d							
Whole reefs																											
John Brewer	18	47	17	19	16	63	10	10	36	14	19	30	17	14	21	49	36	8	14	44	84	19	43	43	45	(36, 12)	
Rib	42	35	13	11	37	63	0	0	29	33	14	24	23	27	17	33	4	16	50	30	72	11	51	60	49	(30, 11)	
Potter	26	37	11	25	37	47	9	7	30	24	25	20	27	31	19	23	36	23	32	7	155	57	83	52	22	(32, 10)	
Feather	13	39	25	25	8	58	25	8	22	34	31	13	15	55	11	18	34	18	20	29	98	48	94	65	56	(36, 16)	
Green Island	31	40	7	21	31	50	4	15	26	33	14	28	34	25	14	26	20	30	9	41	90	54	125	76	56	(33, 21)	
All reefs	26	40	15	20	23	56	10	8	29	28	21	23	23	30	17	30	20	19	25	30	499	189	396	296	228	(167, 70)	
Individual sectors																											
John Brewer																											
(a)	0	0	25	75	0	0	0	0	33	0	67	0	20	20	40	20	22	22	22	33	4	0	3	5	9	(3, 1)	
(b)	0	42	29	29	0	67	0	33	25	0	14	57	8	39	15	39	50	0	0	50	7	3	7	13	8	(3, 1)	
(c)	0	29	0	71	0	0	0	0	33	33	33	0	25	0	0	75	33	0	33	33	7	0	3	4	3	(3, 1)	
(d)	22	33	33	11	0	100	0	0	14	14	42	29	0	0	33	67	20	20	0	60	9	2	9	3	5	(3, 1)	
(e)	77	11	11	0	100	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	9	2	0	0	0	(3, 1)	
(f)	13	63	13	13	0	100	0	0	0	0	0	0	0	0	0	0	0	0	0	0	8	2	0	0	0	(3, 1)	
(g)	0	100	0	0	0	50	0	50	0	20	40	40	0	0	0	100	0	100	0	0	11	2	5	2	1	(3, 1)	
(h)	75	25	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4	0	0	0	0	(3, 1)	
(i)	8	39	8	46	0	0	0	0	50	30	0	20	0	0	20	80	37	0	14	29	13	0	10	5	7	(3, 1)	
(j)	0	50	50	0	0	0	0	0	50	0	0	50	20	0	40	40	0	0	0	100	2	0	4	5	4	(3, 1)	
(k)	0	0	100	0	0	100	0	0	67	0	0	33	67	0	0	33	50	0	25	25	1	1	3	3	4	(3, 1)	
(l)	27	37	37	0	17	67	0	17	0	0	0	0	50	0	0	50	50	0	25	25	11	6	0	2	4	(3, 1)	
Rib																											
(a)	0	30	20	50	0	0	0	0	0	50	0	50	0	36	27	36	9	27	37	27	10	0	2	11	11	(3, 3)	
(b)	14	57	29	0	33	67	0	0	17	17	21	43	6	18	24	53	0	9	57	35	7	3	23	17	23	(3, 3)	
Potter																											
(a)	54	14	8	24	61	31	0	8	22	22	40	17	29	21	29	21	30	25	35	10	63	23	52	28	20	(15, 6)	
Feather																											
(a)	9	50	22	19	6	63	25	6	38	36	14	11	36	50	0	14	67	24	9	0	32	16	44	28	21	(12, 8)	
(b)	17	33	50	0	0	100	0	0	8	33	45	14	0	68	11	21	11	19	22	48	6	1	49	28	27	(6, 6)	
Green Island																											
(a)	67	33	0	0	0	67	0	33	71	29	0	0	100	0	0	0	38	62	0	0	13	3	7	5	8	(3, 4)	
(b)	17	75	8	0	39	61	0	0	6	19	13	62	8	8	42	42	0	14	29	57	12	18	16	12	7	(3, 3)	
(c)	0	29	0	71	0	50	0	50	25	37	19	19	39	27	19	15	38	38	15	8	7	4	52	26	13	(3, 6)	
(d)	25	50	20	0	0	50	50	0	7	29	29	36	0	33	0	67	9	45	9	36	4	4	14	3	11	(3, 3)	

ment of coral size,

$$= (1 - P_1) / ((T - 10) / w), \quad (2)$$

where w = width of the size class in cm, and T = the escape threshold diameter from background mortality in centimeters (in this study, defined as the diameter at which P_i reaches its maximum value of 0.99).

All other $a_{i,j} = 0$, because the growth rate is equal to the size increment.

Values of P_1 and T in the ranges of 0.7 and 200 cm to 0.9 and 60 cm, were found elsewhere to simulate conditions of very severe to very slight background mortality (Done 1987). The abundance and size of colonies in a single cohort are illustrated in Fig. 3.

The disturbance matrices for whole reefs and for specific locations are derived empirically from the field survey data (Table 2), in which the *Porites* spp. populations are subdivided

into Size Classes I to V, with lower class boundaries at 0.1, 0.3, 0.6, 1.0, and 2.0 m. A recruit is defined as a coral entering the population at 0.1 m diam, without regard to its origin as a planula or a fragment.

Disturbance causes shrinkage of corals from Class i to Class j according to:

$$a_{j,j} = \sum_s A_{i,j} \times B_{i,j}, \quad (3)$$

where the population is grouped into the five size classes above, and $A_{i,j}$ = the proportion of Class i which is caused to shrink to Class j by damage level s , and $B_{i,j}$ = the proportion of Class i which suffers damage level s .

Damage levels used in the above were 5/6, 1/2 and 1/6, being midpoints of the damage categories scored in the field. The derivation of the matrices and function of the model are described more fully in Done (1987).

Table 3. *Porites* spp. Population size structure (no. ha⁻¹). (Several numbers appear repeatedly in table as result of using large conversion factors to convert numbers per transect into no. ha⁻¹; see "Materials and methods – Field data")

Location	Size class and (diam range)					Total no. ha ⁻¹
	I (0.1—0.29 m)	II (0.3—0.59 m)	III (0.6—0.99 m)	IV (1.0—1.99 m)	V (≥ 2.0 m)	
Whole reefs						
John Brewer	1 500	185	8	9	12	1 709
Rib	799	122	12	14	11	958
Potter	1 614	593	21	13	6	2 247
Feather	1 815	889	15	10	22	2 751
Green Island	909	545	15	9	7	1 485
Individual sectors						
John Brewer						
(a)	889	0	8	13	23	933
(b)	1 556	667	18	33	20	2 294
(c)	1 556	0	8	10	8	1 582
(d)	2 000	444	18	8	10	2 479
(e)	2 000	444	0	0	0	2 444
(f)	1 778	444	0	0	0	2 222
(g)	2 444	444	12	5	3	2 908
(h)	889	0	0	0	0	889
(i)	2 889	0	25	12	17	2 943
(j)	444	0	10	12	10	476
(k)	222	222	8	8	10	470
(l)	2 444	1 111	0	5	10	3 348
Rib						
(a)	1 111	0	2	9	9	1 131
(b)	777	333	19	14	19	1 162
Potter						
(a)	1 400	511	21	11	8	1 951
Feather						
(a)	888	444	13	8	6	1 359
(b)	333	55	20	12	11	431
Green Island						
(a)	1 667	333	4	3	5	2 012
(b)	1 333	1 999	13	10	5	3 360
(c)	778	444	22	10	5	1 259
(d)	444	444	11	3	9	911

Using the model to evaluate field data

Differences in the severity of disturbance among the five reefs were evaluated by simulating the responses of the *Porites* spp. populations. The initial population vector (viz. the observed field abundances in Size Classes I to V) was multiplied once by the disturbance matrix (producing the post-disturbance size-structure), and then allowed to grow according to a non-disturbance matrix over consecutive years.

Three series of simulations were conducted. The first two followed the fate of only those individuals surviving the original disturbance. As well as circumventing the problem of predicting future recruitment, they provide worst-case scenarios in which the initial disturbance is assumed to cause an irreversible change in the substratum, making it unsuitable for further colonisation. The third series simulated the effects of recruitment on size structure in disturbed populations.

Series 1: "recovery time" simulations (no recruitment)

In these simulations, the time taken for the re-establishment of the pre-disturbance abundance of the largest corals (Class V) was simulated in the absence of further disturbance.

Series 2: "rundown" simulations (no recruitment)

Here, the populations were run down to extinction by repeated disturbances, each identical in character and separated by 10, 20 or 30 yr recovery periods. During these periods, *Porites* spp. were subjected to one of the four regimes of background mortality (Fig. 3). The number of disturbances to extinction is an index of the severity of the disturbance and its present vulnerability.

Series 3: "breakeven" simulations (constant recruitment)

These simulations demonstrated long-term effects of recurrent disturbances on population structure. Recruitment

Table 4. *Porites* spp. Initial population vectors, V (from Table 3), disturbance matrices [A , derived from Table 2 using Eq. (3)], and mortalities (M , being 1.0 minus column totals of A)

John Brewer (whole reef)						John Brewer (j)					
V=	1500	185	8	9	12	V=	444	0	10	12	10
A=	.269	.689	.049	0	0	A=	.535	0	0	0	0
	.052	.127	.097	.067	0		0	0	.070	0	0
	0	.014	.269	.210	.013		0	0	.430	.208	0
	0	0	.024	.560	.134		0	0	0	.592	.080
	0	0	0	.002	.381		0	0	0	0	.920
M=	.679	.170	.561	.161	.472	M=	.465	0	.500	.200	0
John Brewer (a)						John Brewer (k)					
V=	889	0	8	13	23	V=	222	222	8	8	10
A=	.910	0	0	0	0	A=	.790	1	0	0	0
	0	0	.453	.094	.040		0	0	.047	0	0
	0	0	.213	.292	.282		0	0	.286	.037	.102
	0	0	0	.258	.456		0	0	0	.296	.398
	0	0	0	0	0		0	0	0	0	0
M=	.09	0	.333	.356	.222	M=	.210	0	.667	.667	.500
John Brewer (b)						John Brewer (l)					
V=	1556	667	18	33	20	V=	2444	1111	0	5	10
A=	.617	.700	0	0	0	A=	.389	.683	0	0	0
	0	.300	.177	.181	0		0	.150	0	0	0
	0	0	.537	.309	0		0	0	0	.055	0
	0	0	0	.433	.040		0	0	0	.445	.102
	0	0	0	0	.460		0	0	0	0	.398
M=	.383	0	.286	.077	.500	M=	.611	.167	0	.500	.500
John Brewer (c)											
V=	1556	0	8	10	8						
A=	.758	0	.117	0	0						
	0	0	.433	0	0						
	0	0	.107	.083	0						
	0	0	0	.668	.137						
	0	0	0	0	.529						
M=	.241	0	.333	.250	.335						
John Brewer (d)						Rib (whole reef)					
V=	2000	444	18	8	10	V=	799	122	12	14	11
A=	.462	.420	.050	0	0	A=	.305	.630	.116	0	0
	0	0	.424	0	0		0	0	.343	.127	0
	0	0	.383	.210	0		0	0	.251	.249	.029
	0	0	0	.789	.212		0	0	0	.394	.320
	0	0	0	0	.552		0	0	0	0	.611
M=	.538	.580	.143	.001	.236	M=	.695	.370	.290	.230	.040
John Brewer (g)						Rib (a)					
V=	2444	444	12	5	3	V=	1111	0	2	9	9
A=	.280	.550	.070	0	0	A=	.717	0	.175	0	0
	0	.450	.458	0	0		0	0	.395	.171	0.049
	0	0	.472	.110	.180		0	0	.430	.344	.365
	0	0	0	.890	.820		0	0	0	.484	.494
	0	0	0	0	0		0	0	0	0	0
M=	.720	0	0	0	0	M=	.283	0	0	0	.092
John Brewer (i)						Rib (b)					
V=	2889	0	25	12	17	V=	777	333	19	14	19
A=	.607	0	.105	0	0	A=	.386	.667	.061	0	0
	0	0	.223	0	0		0	0	.322	.083	0
	0	0	.172	.170	0		0	0	.433	.248	.015
	0	0	0	.830	.070		0	0	0	.610	.286
	0	0	0	0	.359		0	0	0	0	.699
M=	.393	0	.500	0	.571	M=	.614	.333	.174	.059	0

Table 4 (continued)

Potter (whole reef)						Green Island (whole reef)					
<i>V</i> =	1614	593	21	13	6	<i>V</i> =	909	545	15	9	7
<i>A</i> =	.428	.515	.084	0	0	<i>A</i> =	.357	.532	.116	0	0
	0	.115	.354	.146	0		0	.158	.349	.118	0
	0	0	.267	.268	.041		0	0	.286	.219	.053
	0	0	0	.317	.300		0	0	0	.296	.309
	0	0	0	0	.279		0	0	0	0	.438
<i>M</i> =	.527	.370	.295	.269	.380	<i>M</i> =	.644	.310	.250	.368	.200
Potter (a)						Green Island (a)					
<i>V</i> =	1400	511	21	11	8	<i>V</i> =	1667	333	4	3	5
<i>A</i> =	.329	9.313	.074	.0	0	<i>A</i> =	.093	.7	.1	0	0
	0	.078	.436	.100	0		0	.3	.186	0	0
	0	0	.278	.254	.045		0	0	0	0	.112
	0	0	0	.359	.328		0	0	0	0	.512
	0	0	0	0	.326		0	0	0	0	0
<i>M</i> =	.671	.609	.212	.287	.301	<i>M</i> =	.907	0	.714	0	.376
Feather (whole reef)						Green Island (b)					
<i>V</i> =	1815	889	15	10	22	<i>V</i> =	1333	1999	13	10	5
<i>A</i> =	.544	.693	.119	0	0	<i>A</i> =	.276	.611	.065	.039	0
	0	.217	.450	.259	0		0	0	.294	0.39	0
	0	0	.211	.357	.032		0	0	.577	.261	.026
	0	0	0	.225	.237		0	0	0	.617	.257
	0	0	0	0	.401		0	0	0	0	.717
<i>M</i> =	.456	.090	.220	.160	.033	<i>M</i> =	.724	.389	.064	.083	0
Feather (a)						Green Island (c)					
<i>V</i> =	888	444	13	8	6	<i>V</i> =	778	444	22	10	5
<i>A</i> =	.492	.736	.127	0	0	<i>A</i> =	.759	.550	.128	0	0
	0	202	.345	.235	0		0	.450	.395	.126	0
	0	0	.142	.281	.043		0	0	.227	.238	.069
	0	0	0	.127	.227		0	0	0	.250	.372
	0	0	0	0	0		0	0	0	0	.174
<i>M</i> =	.508	.062	.386	.357	.730	<i>M</i> =	.241	0	.250	.386	.385
Feather (b)						Green Island (d)					
<i>V</i> =	333	55	20	12	11	<i>V</i> =	444	444	11	3	9
<i>A</i> =	.488	1	.114	0	0	<i>A</i> =	.337	.710	.100	0	0
	0	0	.537	.319	0		0	.290	.430	.156	0
	0	0	.267	.427	.033		0	0	.398	.250	.082
	0	0	0	.254	.263		0	0	0	.593	.432
	0	0	0	0	.591		0	0	0	0	.395
<i>M</i> =	.512	0	0.82	0	.113	<i>M</i> =	.663	0	.072	0	.091

refers to the establishment of 0.1 m diam corals, and is set at 160 ha⁻¹ yr⁻¹ in all simulations. This rate is an estimate based on pooled data for Class I corals at John Brewer Reef. The consequences of variable recruitment were investigated by Done (1987).

Results

Coral structure and starfish damage in field populations *Porites* spp. colonies were encountered at a rate of generally less than ten records per transect (Table 2), equivalent to

total densities (Table 3) ranging from ~ 400 ha⁻¹ (at Sector b of Feather Reef) to ~ 3000 ha⁻¹ (at Sector l of John Brewer Reef and Sector b of Green Island Reef) For corals > 1 m diam, the maximum density was 53 ha⁻¹, or approx 1 per 200 m² (at Sector b of John Brewer Reef). There was a general pattern of decreasing numerical abundance with increasing size, typically a range of about two orders of magnitude between the smallest and the largest size classes; in several sectors, larger corals were completely absent (viz. Sectors e, f, and h of John Brewer Reef: Table 3).

No simple or consistent pattern of mortality in relation to coral size was apparent (Table 2). Mortality was greatest

Table 5. *Porites* spp. Projections of recovery times and number of outbreaks to extinction assuming no recruitment and background mortalities as depicted in Fig. 3. Dash indicates failure of stock present in survey to recover pre-disturbance abundance of Class V

	Recovery time (yr)				Outbreaks to extinction				Comments
	low	medium	medium	high	low	medium	medium	high	
Background mortality (P_1 , T)	(0.9, 100)	(0.9, 200)	(0.7, 100)	(0.7, 200)	(0.9, 100)	(0.9, 200)	(0.7, 100)	(0.7, 200)	See Fig. 3 for background mortality curves See Eqs. (1) and (2) in text
Whole reefs									
John Brewer	69	94	—	—	7	4–6	4–5	2–3	Slow recovery; vulnerable ^a
Rib	36	93	—	—	6–9	6	6–7	5	Fastest recovery; least vulnerable
Potter	53	81	—	—	6–7	4–5	3	2–3	Medium recovery time; vulnerable
Feather	74	—	—	—	7–8	4–7	4	3	Slowest recovery
Green Island	56	93	—	—	6–7	4–5	3–4	2	Most vulnerable
Individual sectors									
John Brewer									
(a)	88	—	—	—	4–6	3–5	4–5	2	Long recovery time; vulnerable
(b)	54	—	88	—	12	5–7	5–6	3–4	Medium-long recovery time
(c)	33	37	88	—	5–7	4–5	4–5	3	Medium recovery time; moderately vulnerable
(d)	43	53	89	—	10	6–7	8–9	4	Medium-long recovery time
(g)	24	25	60	—	17	8–13	19	2–4	Medium recovery time; moderately vulnerable
(i)	88	—	92	—	8–9	4–5	4–5	3	Long recovery time – vulnerable
(j)	9	10	9	—	> 20	3–5	> 20	2	Short recovery time; variable vulnerability
(k)	88	—	—	—	4–8	2–7	2–3	1–3	Long recovery time; vulnerable
(l)	74	77	88	—	4–7	3–7	2–3	2–3	Long recovery time; vulnerable
Rib									
(a)	62	89	90	—	6	4–5	5–6	2	Medium-long recovery time; vulnerable
(b)	33	40	89	—	15	11	13	6–9	Medium recovery time, least vulnerable
Potter									
(a)	55	40	89	—	6–8	4	4	2	Medium-long recovery time; vulnerable
Feather									
(a)	73	74	88	—	5–9	3–7	2–3	1–3	Long recovery time; vulnerable
(b)	56	59	95	—	8	5–7	5–6	4	Medium recovery time; moderately vulnerable
Green Island									
(a)	73	75	82	—	2–8	2–6	1–3	1–3	Long recovery time; vulnerable
(b)	16	17	17	19	11	8	10	5–6	Short recovery time, not vulnerable
(c)	59	62	75	—	7–10	3–7	3–4	1–3	Medium-long recovery time; vulnerable
(d)	63	73	78	—	14	5–7	7	3	Medium-long recovery time; moderately vulnerable

^a Vulnerable to additional outbreaks displaying the same characteristics as determined for individual reefs and sectors (Table 4)

among the smallest corals (<0.3 m diam in 14 sectors, but in 4 others, it was greatest in corals > 2 m). The maximum mortality observed in any size class was 90% of corals <0.3 m diam at Sector a of Green Island, while the maximum observed among corals > 2 m diam was 73% at Sector a of Feather Reef. The mean mortalities over all reefs (for smallest to largest size classes, based on 189 to 499 observations) were very consistent; viz. 26%, 23%, 29%, 23% and 20% (Table 2).

Slight to major injury by *Acanthaster planci* was observed in all size classes at all sectors (Table 2), and again there was no clear trend of degree of injury with size. Pooling all data, the greatest proportions of surviving corals were in the least injured category (d) for corals > 1 m diam, and conversely, in the most injured category (b) for the three size classes < 1 m diam. For individual reefs and sectors, the numbers of observations were too small to exhibit any spatial pattern which may exist, and it is difficult to draw any

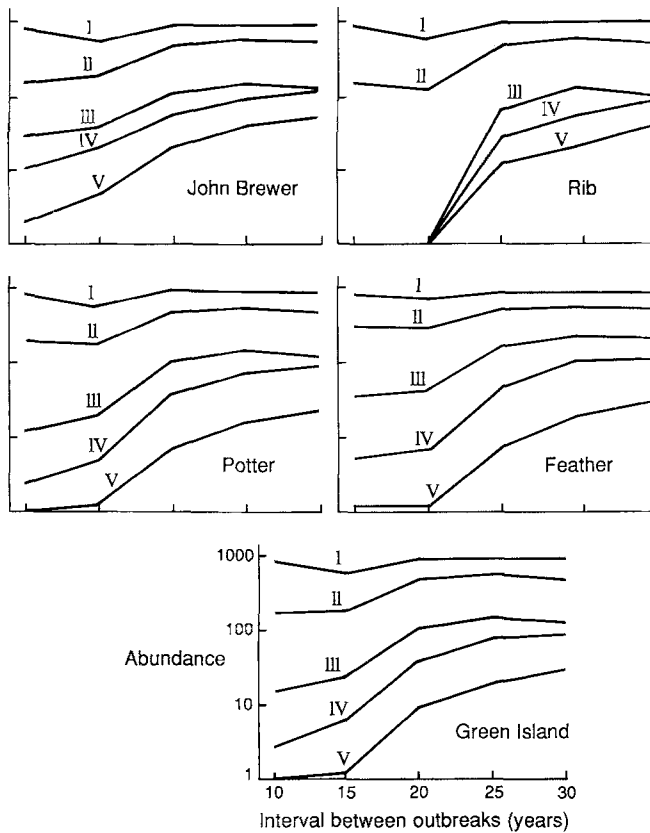


Fig. 4. *Porites* spp. Maximum abundances (no. ha⁻¹) predicted for the five coral size classes after 170 to 200 yr of *Acanthaster planci* outbreaks at 10, 15, 20, 25 and 30 yr. Colonies had a high initial annual probability of survival ($P_1 \approx 0.9$) and a small escape threshold diameter ($T = 100$ cm) – the low background mortality depicted as the upper curve in Fig. 3. Recruitment was 160 ha⁻¹ yr⁻¹ of colonies of 0.1 m diam. Under these conditions, all sizes were able to survive at all reefs when disturbance intervals were long, but larger size classes were lost at four of the five reefs when the interval was 15 yr or less

clear inferences from the raw data. All that is clear is that the high variability in injury and mortality compounds an equally high variability in population size structure (Table 3). The simulations which follow help to clarify the consequences of these variable impacts on variable population structures, which are expressed as 23 disturbance matrices and population vectors in Table 4.

“Recovery” simulations

Under a regime of low background mortality, all populations of *Porites* spp. recovered pre-disturbance abundances of Class V corals within 88 yr (Column 2 in Table 5). The projected recovery times for whole reef populations ranged from 36 yr (Rib Reef) to 74 yr (Feather Reef). Recovery times for individual sectors ranged from 9 to 88 yr.

However, with high background mortality (Column 5), only Green (b) population recovered, and it did so relatively quickly (19 yr). The remaining populations failed to recover because background mortality depleted populations excessively. These populations would depend for recovery on

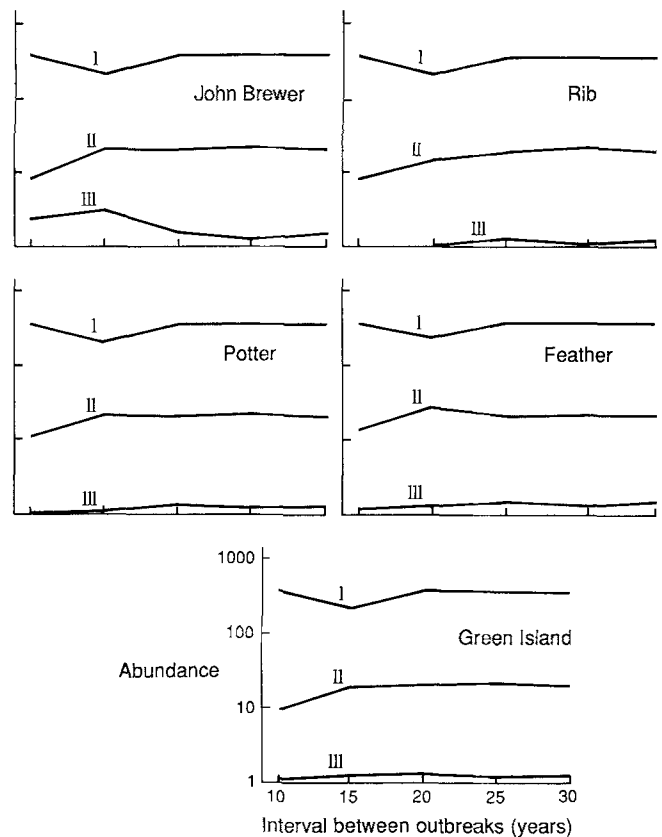


Fig. 5. *Porites* spp. Predicted maximum abundances (no. ha⁻¹) for same coral populations as in Fig. 4. In this case, because of high background mortality ($P_1 = 0.7$, $T = 200$ cm – lowest curve in Fig. 3), the larger corals failed to survive at any disturbance interval up to and including 30 yr

the establishment of new colonies, as would all others in Table 5 for which no recovery time is indicated. These new colonies would take about 100 yr to reach a diameter of 2 m.

“Rundown” simulations

The number of additional disturbances needed to drive the survivors of the initial disturbance to extinction (Table 5) is an indication of both the severity of the disturbance and the vulnerability of the survivors to further disturbances of the same intensity. By this criterion, Rib Reef was again shown to be the least severely affected, i.e., would require the greatest number of additional identical disturbances to cause extinction of the population. This number is five in the case of high background mortality (Column 9 in Table 5). By contrast, John Brewer Reef and Green Island could withstand only two additional disturbances. Values for individual sites range between 1 and 9 disturbances (high background mortality) or 2 and > 20 (low background mortality).

“Breakeven” simulations

When 0.1 m diam recruits were introduced each year, the population could maintain some, or all of its size classes indefinitely, even when confronted with regular disturbance.

At all five reefs, a best-case scenario involving constant high recruitment combined with low background mortality allowed the larger corals to reinstate their pre-disturbance abundances at some interval < 30 yr (Fig. 4). By contrast, in worst-case scenarios (either poor recruitment, high background mortality, or both; Fig. 5) the populations permanently lost their larger corals.

The best case simulations appear to fall into three groups: (1) Potter, Feather and Green Reefs (in which all Class V colonies are lost at intervals < 15 yr); (2) John Brewer Reef (in which some Class V colonies are retained at all intervals of 10 to 30 yr); (3) Rib Reef (in which Classes III to V are lost at intervals < 20 yr.).

These simulations also indicate how abundances of small corals (Classes I and II) are, not surprisingly, controlled by the constant, high recruitment to the reefs. Their long-term abundances are almost identical, whatever the disturbance interval, indicating that they are essentially unaffected by differences in the disturbances.

Discussion

Porites spp. populations on the slopes of these reefs are of mixed size frequencies, and tend to be dominant numerically by small individuals. They are preyed upon by outbreaking populations of the crown of thorns starfish *Acanthaster planci*, and all sizes suffer some degree of injury and mortality. If we assume recovery to be complete only when the abundance of the largest colonies is restored, then recovery times of decades to a century are indicated in most places, assuming low background mortality and no further disturbances. This estimate is much shorter than the age of the largest corals, because replacement results from repair of injured colonies and growth from intermediate size classes rather than from new recruits. Hence, the mixed size structure of the population can provide a buffer which does not exist in populations lacking large numbers of small and intermediate-sized colonies (e.g. Potts et al. 1985). Replacement of the small and intermediate size classes through recruitment is shown to be critical in determining whether large corals may exist in the long term in a regularly disturbed population.

The analysis suggests that *Porites* spp. populations on these starfish-affected reefs are presently in a vulnerable state. The total *Porites* spp. population at Green Island could withstand as few as two additional outbreaks of the intensity recently documented, given high but plausible background mortality. Small residual populations of starfish faced with limited supplies of coral may cause a major component of this background mortality. Under the same circumstances, even the least disturbed of the reefs surveyed (Rib) could withstand only five more disturbances. At specific locations on the various reefs, 1 to > 20 disturbances of the same intensity would make the local population extinct.

Moreover, even under favourable conditions, most of the reefs would take at least 50 yr to establish their previous

abundances of corals > 2 m diam. For specific locations, recovery times range from 9 to > 100 yr, with the great majority > 50 yr. Recovery times for larger corals (say > 3 m diam) would be up to 50 yr greater than for the estimates for > 2 m corals. These estimates refer to recovery of population densities and size structures in the pre-1980's which were possible to estimate following the outbreaks, since dead *Porites* spp. remain recognisable for several years. However, no comparable data exist for populations of *Porites* spp. or any other coral prior to the outbreaks in the 1960's.

High levels of spatial variability, both in the coral populations, in the impact of the starfish and in projected recovery times, are a notable feature of this analysis. No single observation provided an adequate sample of an entire reef, and no reef was representative of the entire region. Regional generalizations are therefore risky. For example, while it is safe to predict that none of the whole-reef populations will recover their pre-disturbance abundances of large corals in less than 36 yr (Table 5), it is equally true that one reef will take more than twice as long if the same low background mortality is assumed. Some populations will take at least a century if background mortality depletes the survivors of the original outbreak. Some populations which were little affected should recover within a decade.

Conclusions

This study indicates that variability in both the population structure of *Porites* spp. and damage by the crown of thorns starfish *Acanthaster planci* is considerable in the case of both whole-reef and localised populations. Recovery time depends on these variables as well as on the criterion for recovery. When re-establishment of pre-disturbance abundance of large corals is the criterion for recovery, recovery times are decades to centuries. These periods contrast with the much shorter times necessary for re-establishment of a pre-existing level of coral cover by faster growing species (Pearson 1981) and with the much longer times necessary for complete replacement of the oldest corals (Cameron and Endean 1985). Indices of disturbance which recognise coral population and community structure as attributes are more appropriate than simple coral cover, the shortcomings of which have been noted elsewhere (e.g. Johannes 1975, Pearson 1981, Endean and Cameron 1985).

The analysis indicates that recurrent outbreaks could have occurred periodically but it suggests that the recently observed interval of about 15 yr is unlikely to be sustainable (Fig. 4). The persistence in a disturbed population of a mixed size structure including large corals, relies on both a reliable recruitment and low background mortality, e.g. in the model population, survival of about one in ten 0.1 m corals to a size of 2.0 m diam (Fig. 3). Except in the case of John Brewer Reef, a 15 yr interval between outbreaks of the same intensities as those just passed, would result in the loss of large colonies from the population, in spite of reliable recruitment and low background mortality. In another sim-

ulation study, it was shown that even the John Brewer Reef population would not persist in the face of 15-yearly outbreaks, under many plausible scenarios of recruitment, survivorship and colony growth rate (Done 1987).

The present study has implications for conservation management of these slow-growing corals. First, since recruitment rate can determine whether a population can retain any large corals at all in the presence of recurrent disturbance, the husbandry of natural recruitment and the introduction of small corals are sound long-term options, although impractical except over small areas. The timing of recruitment between outbreaks, be it episodic or regular, does not significantly influence the long-term (hundreds of years) abundance of large corals in disturbed populations (Done 1987).

Second, controlling further background mortality would be beneficial, because it influences recovery time and vulnerability to further disturbance. In principle this requires the maintenance of a good reef environment, and in practice the minimization of controllable deleterious inputs, i.e., those which are outside the limits of accommodation by the coral reef ecosystem. These limits are only poorly understood at both physiological and ecological levels and warrant clearer definition by managers and scientists.

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